PROOFLEDGER | Explainable Tax Workpaper

MiraFlores AI - 2025 Okafor-Lindqvist fictional return | Architecture + data model | Case-study prototype

Core idea - an evidence graph, not a spreadsheet

Every computed return line points backward to a calculation, source document and exact field, then forward to verification, exception, adjustment, reviewer judgment and approval. The UI prioritizes unresolved proof rather than displaying thousands of cells. The supplied tax engine remains authoritative for complex tax results; arithmetic and cross-source reconciliations run independently.

System architecture

Client + CPA interfaces	Server / persistence	External boundary
Evidence library; return explorer; case workspace; client portal	Python HTTP API; staff / client sessions; role + case authorization; SQLite with revision checks	Supplied JSON packet; scripted IRS/FTB timeline; no actual e-filing
Calculation lab; exceptions; reviews; filing control; audit receipt	Immutable filing checkpoint; approval fingerprints; encrypted new case evidence; event and view logs	Qualified e-signature and identity proofing require real providers

Data model (key entities and relationships)

Document 1-to-many ExtractedField - linked via LineSource to ReturnLine and Calculation. ReviewDecision and Adjustment reference evidence; StaffSignoff and ClientAcknowledgment reference a canonical review fingerprint; FilingEvent and FrozenCheckpoint preserve the accepted version. FirmCase has per-case Membership, Task, TimeEntry, ReviewRecord and EncryptedEvidence.

People, filing, firm: working flows

People	Client-specific expiring invitations, independent decisions, questions as durable evidence, revision invalidation; optional individual staff passwords + TOTP.
Filing	Guarded fixture-rejection/acceptance lifecycle, frozen checkpoint, separate amendment ledger, notice evidence bundle and SHA-256 review-authorization receipt.
Firm	Searchable cases, restricted memberships, tasks, time vs budget, decisions, encrypted new-case attachments with verified checksum and audited retrieval.

Boundaries and integrity strategy

The source packet provides no rerunnable tax engine, IRS credential or signature service. We do not fabricate tax results, government transmissions or legally valid Form 8879 signatures. New uploaded evidence is encrypted, but the supplied packet and other review data are not encrypted; this is not production security. Additional firm cases are working operational workspaces, not fictitious completed returns.

Tests: 101 HTTP assertions, 8 packet checks, Chromium UI smoke tests, and syntax checks.